

Harmonic Run-and-Tumble Dynamics: Beta Stationarity, Correlations, and Polynomial Jordan Resonances

Route-A source-local certificate HCS-C328

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Abstract

We close the invariant law of a one-dimensional harmonic run-and-tumble particle, including its two orientation components and every polynomial moment.

1 Stationary beta law and all moments

Let $\sigma_t \in \{-1, 1\}$ flip to its opposite at rate λ , and let

$$dx_t = (v\sigma_t - \mu x_t) dt, \quad \mu, v, \lambda > 0. \quad (1)$$

The generator is

$$Lf_\sigma = (v\sigma - \mu x)\partial_x f_\sigma + \lambda(f_{-\sigma} - f_\sigma).$$

Set $a = v/\mu$, $y = x/a$, and $\alpha = \lambda/\mu$.

Theorem 1 (Stationary components and moments). *The process has a unique stationary probability on $[-a, a] \times \{-1, 1\}$. With respect to dy , its marginal and joint component densities are*

$$p(y) = \frac{\Gamma(\alpha + \frac{1}{2})}{\sqrt{\pi} \Gamma(\alpha)} (1 - y^2)^{\alpha-1}, \quad (2)$$

$$p_+(y) = \frac{1+y}{2} p(y), \quad p_-(y) = \frac{1-y}{2} p(y). \quad (3)$$

For every $n \geq 0$,

$$\mathbb{E}x^{2n} = a^{2n} \frac{(1/2)_n}{(\alpha + 1/2)_n}, \quad \mathbb{E}x^{2n+1} = 0, \quad (4)$$

$$\mathbb{E}(\sigma x^{2n}) = 0, \quad \mathbb{E}(\sigma x^{2n+1}) = a^{2n+1} \frac{(1/2)_{n+1}}{(\alpha + 1/2)_{n+1}}. \quad (5)$$

Proof. The deterministic vector fields point inward at $\pm a$. In stationarity, adding the two forward equations makes total probability flux constant; the no-flux boundary gives $(1-y)p_+ - (1+y)p_- = 0$. Hence (3) holds. Substitution in either remaining equation gives $p'/p = -2(\alpha-1)y/(1-y^2)$, and beta normalization gives (2). Positive flip rate makes the compact process irreducible in its accessible interior, giving uniqueness. Symmetry gives odd moments zero. Finally, integration by parts applied to $y^{2n+1}(1-y^2)^\alpha$ yields

$$\frac{\mathbb{E}y^{2n+2}}{\mathbb{E}y^{2n}} = \frac{n+1/2}{n+\alpha+1/2},$$

The component imbalance in (3) is $yp(y)$, so parity gives every even mixed moment zero and the displayed odd mixed moments. $\square$

Source lineage

References

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- [2] R. Garcia-Millan and G. Pruessner, “Run-and-tumble motion in a harmonic potential: field theory and entropy production,” *J. Stat. Mech.* (2021), 063203. DOI: [10.1088/1742-5468/ac014d](https://doi.org/10.1088/1742-5468/ac014d).